

Module 6: The ALL Function

6.1 Concept

The `ALL` function **removes filters** from a table or column, allowing calculations **ignoring current context**. It's commonly used for:

- Calculating **percent of total**
- Creating **rankings**
- Comparing a subset to the full dataset

Syntax:

```
ALL(tableOrColumn)
```

6.2 Practical Examples

Example 1: Percent of Total Sales

```
% of Total Sales :=  
DIVIDE(  
    SUM(Sales[SalesAmount]),  
    CALCULATE(SUM(Sales[SalesAmount]), ALL(Sales))  
)
```

- ◆ Calculates each row's share of overall sales, ignoring filters.

Example 2: Percent of Category Sales

```
% of Category Sales :=  
DIVIDE(  
    SUM(Sales[SalesAmount]),  
    CALCULATE(SUM(Sales[SalesAmount]), ALL(Sales[Product]))  
)
```

- ◆ Shows each product's contribution to its category.

Example 3: Rank Products by Sales

```
Product Sales Rank :=  
RANKX(  
    ALL(Sales[Product]),  
    SUM(Sales[SalesAmount]),  
    ,  
    DESC  
)
```

- ◆ Assigns rank to each product based on total sales, ignoring filters.

Example 4: Category Rank by Profit

```
Category Profit Rank :=  
RANKX(  
    ALL(Sales[Category]),  
    SUM(Sales[Profit]),  
    ,  
    DESC  
)
```

- ◆ Ranks categories by total profit.

6.3 Exercises (Hands-On)

1. Percent of Total Profit

- Measure: Each product's profit as % of **all product profits**.

2. Product Contribution within Category

- Measure: Each product's `SalesAmount` as % of total for its **category**.

3. Top Product by Sales Rank

- Measure: Rank all products by `SalesAmount` in descending order.

4. Category Rank by Total Sales

- Measure: Rank categories by total `SalesAmount` .

5. Ignore Category Filter

- Measure: Total Sales ignoring any **category slicers**.

6.4 Visualizations in Power BI

1. Table

- Product | `SalesAmount` | % of Total Sales | Product Sales Rank

2. Matrix

- Category | Product | % of Category Sales | Product Sales Rank

3. Bar Chart

- Categories ranked by profit using `Category Profit Rank`

6.5 Exercise Answers (For Trainers)

1. Percent of Total Profit

```
% Total Profit :=  
DIVIDE(  
    SUM(Sales[Profit]),  
    CALCULATE(SUM(Sales[Profit]), ALL(Sales))  
)
```

- Shows each product's contribution to overall profit.

2. Product Contribution within Category

```
% of Category Sales :=  
DIVIDE(  
    SUM(Sales[SalesAmount]),  
    CALCULATE(SUM(Sales[SalesAmount]), ALL(Sales[Product]))  
)
```

- Within a category, each product shows its share of sales.

3. Top Product by Sales Rank

```
Product Sales Rank :=  
RANKX(  
    ALL(Sales[Product]),  
    SUM(Sales[SalesAmount]),  
    ,  
    DESC  
)
```

- Highest-selling product → Rank 1

4. Category Rank by Total Sales

```
Category Sales Rank :=  
RANKX(  
    ALL(Sales[Category]),  
    SUM(Sales[SalesAmount]),  
    ,  
    DESC  
)
```

- Electronics usually rank highest, followed by Accessories/Fashion.

5. Ignore Category Filter

```
Total Sales Ignoring Category :=  
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    ALL(Sales[Category])  
)
```

- Returns overall sales total even when a category slicer is applied.

✅ By the end of Module 6, students will:

- Use `ALL` to **ignore filters** in calculations.
- Calculate **percent of total** for products or categories.
- Create **rankings** for products and categories.
- Build **dynamic, filter-resistant measures** for dashboards.